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SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
IACUC: The TCP animal care committee approved all experimental procedures.
IRB: Human B-ALL samples The Research Ethics Boards at the Hospital for Sick Children (Toronto, Canada) and UHN (Toronto, Canada) approved this study.
Consent: Bone marrow or peripheral blood samples from newly diagnosed patients with ALL were obtained with informed consent.
Euthanasia Agents: 17 , 53 Briefly, mice were anesthetized by isoflurane inhalation, and the area of the injection site just below the patella or femur kneecap was cleaned using a sterile alcohol wipe.
Inclusion and Exclusion Criteria
not detected.
Attrition
not detected.
Sex as a biological variable
TM and nonleukemic SCID male or female mice (8 to 12 weeks old) were used in the experiments.
Subject Demographics
Age: TM and nonleukemic SCID male or female mice (8 to 12 weeks old) were used in the experiments.
Randomization
For PDX experiments, mice were randomly assigned to the IgG1-Fc control or rOPG-Fc or AMD3100 treatment groups.
Blinding
not detected.
Power Analysis
not detected.

Replication
not required.
Cell Line Authentication
not detected.

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies			
STARMethods Key resources table REAGENT or RESOURCE SOURCE IDENTIFIER Antibodies CD3 (UCHT1) BioLegend Cat 300430 RRID:AB_893301 CD45 (2D1) BD Biosciences Cat 347463 RRID:AB_400306 CD19 (4G7) BD Biosciences Cat 349209 RRID:AB_400407 RANKL (12A668) Abcam Cat ab45039 RRID:AB_2205935 CD19 (EPR5906) Abcam Cat ab134114 RRID:AB_2801636 CD19 (EP169) BioGenex Cat AN729-5M CD19 (EPR23174-145) Abcam Cat ab245235 RRID:AB_2895109 CXCR4 (UMB2) Abcam Cat ab124824 RRID:AB_10975635 Biotinylated secondary antibody, Mouse and Rabbit SS Link-label IHC detection system BioGenex Cat QP900-LE (HK340-9KT-multi link) Biotinylated secondary antibody, Mouse and Rabbit HRPDAB (ABC) detection kit Abcam Cat ab64264 CXCR4 (12G5) BD Biosciences Cat 560669 RRID:AB_1727435 Biotinylated rOPG-Fc (in-house) Biotinylated in-house using EZ-link sulpho-NHS biotin kit rOPG-Fc: Amgen and NHS-Biotin kit: Thermoscientific Cat 21326 (NHS Biotin Kit) Streptavidin-PerCP-Cy5.5 e-Bioscience (Thermo-Fisher Scientific) Cat 45-4317-82 RRID:AB_10311495 Biological samples Patient-Derived B-ALL Samples Table S1 Table S1 Chemicals, peptides, and recombinant proteins Recombinant human OPG-Fc protein (RANKL antagonist) Amgen Amgen IgG1-Fc protein (produced in-house using pFUSE-Fc plasmid) Invivogen Catalog pfuse-hg1fc2 plasmid construct AMD3100 octahydrochloride (CXCR4 antagonist) Sigma-Aldrich A5602-5MG Experimental models: organismsstrains Mouse: p53--Rag2-- Prkdcscidscid triple-mutant (TM) The center for Phenogenomics	CD3	BD Biosciences	(BD Biosciences Cat# 349209, RRID:AB_400407)(link)
	UCHT1		
	REAGENT	BioLegend	(BioLegend Cat# 300429, RRID:AB_893301)(link)
	CD45 2D1	BD Biosciences	(BD Biosciences Cat# 347463, RRID:AB_400306)(link)
	RANKL	Abcam	(Abcam Cat# ab45039, RRID:AB_2205935)(link)
	CD19		
	EPR5906	Abcam	(Abcam Cat# ab134114, RRID:AB_2801636)(link)
	EP169		
	EPR23174-145		
	Cat AN729-5M CD19 EPR23174-145	Abcam	(Abcam Cat# ab245235, RRID:AB_2895109)(link)
	UMB2		
	CXCR4 UMB2	Abcam	(Abcam Cat# ab124824, RRID:AB_10975635)(link)
	HK340-9KT-multi link		
			(BD Biosciences Cat# 560669, RRID:AB_1727435)(link)
	NHS		
	Cat	Thermo Fisher Scientific	(Thermo Fisher Scientific Cat# 45-4317-82, RRID:AB_10311495)(link)
	OPG-Fc protein (RANKL antagonist) Amgen Amgen IgG1-Fc protein		

(TCP) (Toronto, Canada) housed and bred the TM mice.			
Moreover, single-cell suspensions isolated from the bone marrow of BCR-ABL1 PDX mice that were treated with IgG1-Fc control were stained with human-specific cell-surface antibodies for CXCR4-PECy7 (1:150 dilution, clone: 12G5, BD Biosciences), CD45-FITC (1:30 dilution, clone: 2D1, BD Biosciences), CD19-PE (1:30 dilution, clone: 4G7, BD Biosciences).	CD19-PE		
In addition, isolated lymphoblasts from the CNS of MLLr PDX mice after treatment with IgG1-Fc control were stained with antibodies specific for cell-surface CD45-FITC (1:30 dilution, clone: 2D1, BD Biosciences), CD19-PE (1:30 dilution, clone: 4G7, BD Biosciences), CXCR4-PECy7 (1:150 dilution, clone: 12G5, BD Biosciences), and biotinylated rOPG-Fc+Streptavidin-PerCP-Cy5.5 (to detect RANKL) as previously described.	CD45-FITC		
	CXCR4-PECy7		
Slides were incubated in pre-diluted ready-to-use anti-human CD19 (clone: EP169) rabbit monoclonal antibody (Biogenex, USA) overnight at 4degC according to the manufacturers instruction.	anti-human CD19		
After washing, slides were incubated with pre- diluted ready-to-use biotinylated anti-rabbit secondary antibody optimized for Biogenex primary antibodies (SS anti-mouse and rabbit, HK340-9KT-multi-link, Biogenex, USA) for 30 min at RT.	anti-mouse		
	HK340-9KT-multi-link		
As an experimental control to detect non-specific secondary antibody-binding, tissue sections were incubated with pre-diluted ready-to-use biotinylated anti-rabbit secondary antibody without the addition of the primary antibody for 30 min at RT.	anti-rabbit		
In addition, bone sections from BCR-ABL1 PDX mice were stained with anti-mouse or human RANKL (clone: 12A668 1:25 dilution Abcam, ab45039) primary antibody and processed as previously described for human CD19 staining.	human RANKL (clone: 12A668 1:25 dilution Abcam, ab45039) primary antibody		
	human		

17 Dewaxed and rehydrated paraffin-embedded bone sections (as described above in the bone histology section) from TM and nonleukemic SCID mice were stained with anti-mouse CD19 (clone: EPR23174-145 1:750 dilution Abcam, ab245235) or anti-mouse or human RANKL (clone: 12A668 1:25 dilution Abcam, ab45039) primary antibodies and processed as previously described for human CD19 staining.	anti-mouse CD19		
17 Bone sections from MLLr PDX mice were stained with anti-human CD19 (clone: EPR5906 1:250 dilution Abcam, ab134114) or anti-mouse or human RANKL (clone: 12A668 1:25 dilution Abcam, ab45039) primary antibodies and processed as previously described for human CD19 staining.	anti-mouse or human RANKL		
17 Bone sections from MLLr PDX mice were stained with an anti-human CXCR4 (clone: UMB2) primary antibody (1:250 dilution Abcam, ab124824), using the same method as previously described for human CD19 staining.	anti-human CXCR4		
17 The anti-human CXCR4 antibody recognizes the non-phosphorylated C terminus of CXCR4.	CXCR4		
Experimental Models: Cell Lines			
NA Software and algorithms GraphPad Prism https:www.graphpad.com Version 8.4.1, for Mac OS X FlowJo v.10 (Tree Star, Ashland, OR) https:www.flowjo.com Version 10 (Tree Star, Ashland, OR) NDP.view2 https:www.hamamatsu.comusenproducttypeU12388-01index.html Version-U12388-01 QuPath (Quantitative Pathology and Bioimage Analysis) https:qupath.github.io Version 0.2.3 Resource availability Lead contact Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Dr. Jayne S.	Version-U12388-01		
Mice were sacrificed at eight weeks or four and six weeks after leukemic blast injection and treatment for BCR-ABL1 and MLLr PDX mice, respectively.	BCR-ABL1		

Experimental Models: Organisms/Strains			
NA Mouse: NOD.PrkdcscidscidIl2rg tm1WjlSzJ (NSG) The University Health Network (UHN), Max bell-animal resource center (Toronto, Canada), housed and bred the NSG mice in pathogen-free (SPF) conditions.	NOD.PrkdcscidscidIl2rg tm1WjlSzJ		
The UHN animal care committee approved all experimental procedures with the NSG mice.	NSG		
Recombinant DNA			
STARMethods Key resources table REAGENT or RESOURCE SOURCE IDENTIFIER Antibodies CD3 (UCHT1) BioLegend Cat 300430 RRID:AB_893301 CD45 (2D1) BD Biosciences Cat 347463 RRID:AB_400306 CD19 (4G7) BD Biosciences Cat 349209 RRID:AB_400407 RANKL (12A668) Abcam Cat ab45039 RRID:AB_2205935 CD19 (EPR5906) Abcam Cat ab134114 RRID:AB_2801636 CD19 (EP169) BioGenex Cat AN729-5M CD19 (EPR23174-145) Abcam Cat ab245235 RRID:AB_2895109 CXCR4 (UMB2) Abcam Cat ab124824 RRID:AB_10975635 Biotinylated secondary antibody, Mouse and Rabbit SS Link-label IHC detection system BioGenex Cat QP900-LE (HK340-9KT-multi link) Biotinylated secondary antibody, Mouse and Rabbit HRPDAB (ABC) detection kit Abcam Cat ab64264 CXCR4 (12G5) BD Biosciences Cat 560669 RRID:AB_1727435 Biotinylated rOPG-Fc (in-house) Biotinylated in-house using EZ-link sulpho-NHS biotin kit rOPG-Fc: Amgen and NHS-Biotin kit: Thermoscientific Cat 21326 (NHS Biotin Kit) Streptavidin-PerCP-Cy5.5 e-Bioscience (Thermo-Fisher Scientific) Cat 45-4317-82 RRID:AB_10311495 Biological samples Patient-Derived B-ALL Samples Table S1 Table S1 Chemicals, peptides, and recombinant proteins Recombinant human OPG-Fc protein (RANKL antagonist) Amgen Amgen IgG1-Fc protein (produced in-house using pFUSE-	pFUSE-Fc		

Fc plasmid) Invivogen Catalog pfuse-hg1fc2 plasmid construct AMD3100 octahydrochloride (CXCR4 antagonist) Sigma-Aldrich A5602-5MG Experimental models: organismsstrains Mouse: p53--Rag2-- Prkdcscidscid triple-mutant (TM) The center for Phenogenomics (TCP) (Toronto, Canada) housed and bred the TM mice.			
Software and Algorithms			
NA Software and algorithms GraphPad Prism https:www.graphpad.com Version 8.4.1, for Mac OS X FlowJo v.10 (Tree Star, Ashland, OR) https:www.flowjo.com Version 10 (Tree Star, Ashland, OR) NDP.view2 https:www.hamamatsu.comusenproducttypeU12388-01index.html Version-U12388-01 QuPath (Quantitative Pathology and Bioimage Analysis) https:qupath.github.io Version 0.2.3 Resource availability Lead contact Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Dr. Jayne S.	https:www.graphpad.com FlowJo		Suggestion: FlowJo, RRID:SCR_008520(link)
Staining media (SM) was prepared by adding 10mM HEPES (pH 7.2) and 2 calf serum (Wisent Inc.) to Hanks balanced salt solution (GIBCO).	GIBCO		
Cells were stained with human-specific cell-surface antibodies for CD45-FITC (1:30 dilution, clone: 2D1, BD Biosciences), CD19-PE (1:30 dilution, clone: 4G7, BD Biosciences), along with Human TruStain Fc-receptor blocking solution BioLegend and incubated for 30 min at RT.	BD Biosciences		
For PDX experiments, primary BCR-ABL1 B-ALL or NSG-passaged MLLr B-ALL blasts-injected NSG mice were treated with IgG1-Fc control (in-house protein) or rOPG-Fc (Amgen) or AMD3100 octahydrochloride (Sigma) or a combination of rOPG-Fc and AMD3100 at the time of leukemic blast injections.	Sigma		Suggestion: sigma-tau (RRID:SCR_000488)(link)
All data were analyzed using GraphPad Prism software, version 8.4.1, for Mac OS X.	GraphPad Prism		

Other Entities Detected

Your Sentences	Recognized Entity
Statistical Tests	
The two-group analysis was performed using a two-tailed unpaired Students t test with or without Welchs correction for heteroscedasticity.	t test
One-way analysis of variance with Bonferronis or Dunns correction was used to evaluate differences among multiple comparisons.	analysis of variance
	Bonferronis
	Dunns correction

SciScore is an automated tool that is designed to assist expert reviewers by finding and presenting formulaic information scattered throughout a paper in a standard, easy to digest format. ***SciScore is not a substitute for expert review.*** SciScore also checks for the presence and correctness of several unique identifiers, including RRIDs (research resource identifiers) in the manuscript, detects sentences that appear to be missing RRIDs, and can even suggest RRIDs under certain circumstances. **All RRID suggestions should be verified;** only the author can know whether the suggestions are correct.

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